
Software Requirements Specification

For

Paperless Scholarship Disbursement System for PMSSS

Version 1.0 | IEEE 830 Compliant | Approved

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1. Introduction

1.1 Purpose

This SRS defines the functional and non-functional requirements for the Paperless Scholarship Disbursement System for PMSSS. It is intended for system architects, developers, testers, and stakeholders. The system digitizes scholarship workflows including application submission, document verification, approval, tracking, and Direct Benefit Transfer (DBT).

1.2 Scope

The system provides a secure, transparent, and efficient platform for managing PMSSS scholarships. Goals include:

- Automating application submission and verification
- Providing real-time tracking and notifications
- Enabling secure DBT fund transfers
- Reducing paperwork and fraud
- Supporting scalability for future government schemes

1.3 Definitions, Acronyms, and Abbreviations

- PMSSS – Prime Minister’s Special Scholarship Scheme
- DBT – Direct Benefit Transfer
- JWT – JSON Web Token (authentication mechanism)
- UI – User Interface
- DBMS – Database Management System

1.4 References

- IEEE Std 830-1998: Software Requirements Specification
- React Documentation
- Spring Boot Documentation
- MongoDB Documentation
- Research papers on e-Governance systems

1.5 Document Overview

Sections cover system description, features, interfaces, models, non-functional requirements, and future enhancements.

2. Overall Description

2.1 Product Perspective

The system is a web-based application integrated with PMSSS workflows. It interacts with:

- Student portal (application submission)
- Admin dashboard (verification & approval)
- DBT payment gateway (fund disbursement)
- MongoDB database (data storage)

2.2 Product Functions

Function ID	Function Name	Description
F-01	User Authentication	Secure login via JWT
F-02	Student Registration	Account creation with personal & academic details
F-03	Scholarship Application	Multi-step online form
F-04	Document Upload	Secure upload & digital verification
F-05	Application Tracking	Real-time status updates
F-06	DBT Transfer	Secure fund disbursement
F-07	Notifications	Alerts for status changes
F-08	Admin Dashboard	Manage applications & approvals
F-09	Reporting	Generate analytics reports

3. System Features

Each feature will have functional requirements (FR-1.1, FR-1.2, etc.), just like the AI SRS. Example:

Feature: Document Upload & Verification

- FR-4.1: System shall allow students to upload required documents (PDF/JPEG).
- FR-4.2: System shall validate file format and size.
- FR-4.3: Admin shall verify documents digitally.
- FR-4.4: System shall log verification events.

4. External Interface Requirements

4.1 User Interfaces

- Web-based portal accessible via modern browsers (Chrome, Firefox, Edge).
- Student dashboard with modules for registration, application submission, document upload, and status tracking.
- Administrator dashboard for document verification, approval, fund disbursement, and reporting.
- Notification panel for alerts (application status, payment confirmation).

4.2 Hardware Interfaces

- Standard student and administrator workstations with internet access.
- Server infrastructure for hosting backend services (Spring Boot).
- Database server for MongoDB storage.
- Secure payment gateway integration for DBT.

4.3 Software Interfaces

Software Component	Interface Description
React (Frontend)	Provides responsive UI for students and administrators.
Spring Boot (Backend)	RESTful APIs for business logic and workflows.
MongoDB	Database for storing applications, documents, and user data.
JWT Authentication	Security mechanism for login and authorization.
DBT Gateway	API integration for secure fund transfer.

4.4 Communication Interfaces

- **HTTP/HTTPS** – For web portal and API communication.
- **WebSocket** – For real-time notifications and status updates.
- **TLS 1.2 or higher** – For encrypted data transmission.

5. System Models

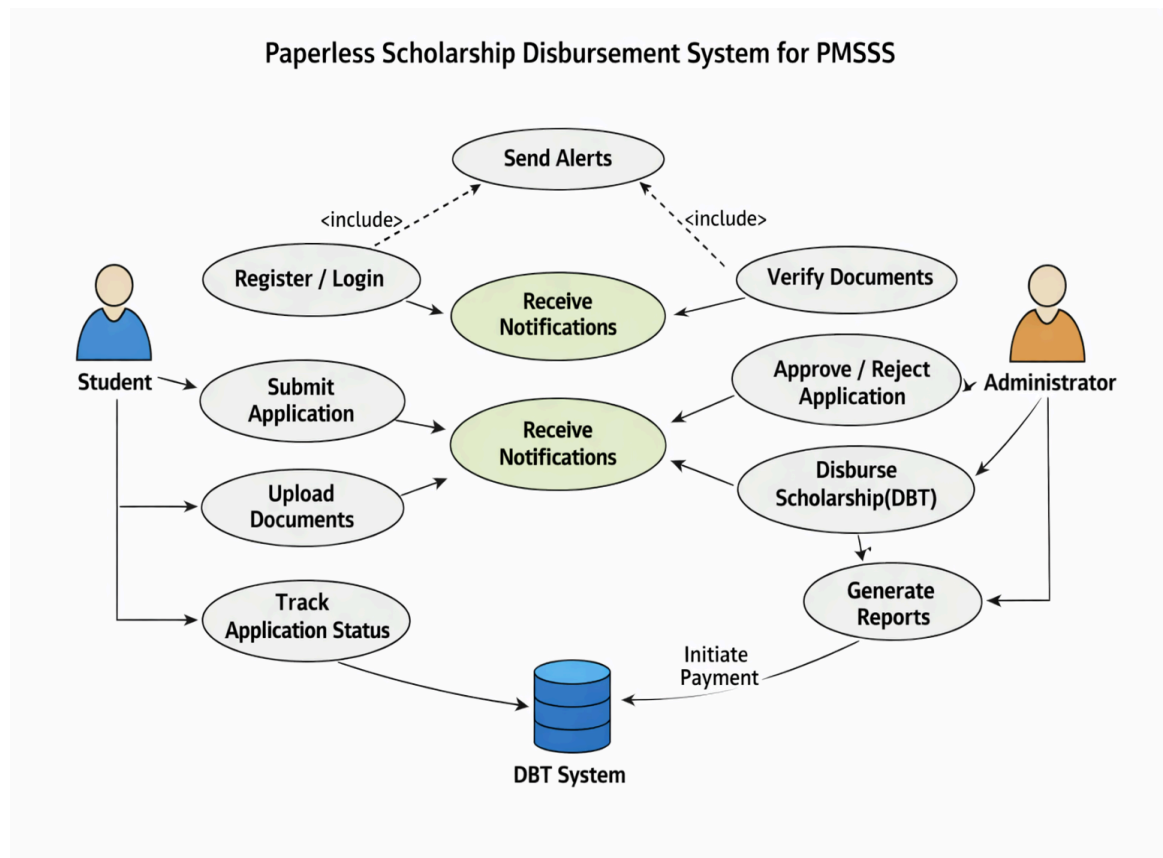
5.1 Use Case Diagram

Actors:

- **Student:** Register, apply, upload documents, track status.
- **Administrator:** Verify documents, approve/reject applications, manage DBT.
- **System:** Automates workflows, sends notifications, manages DBT.

Use Cases:

- Register/Login
- Submit Application
- Upload Documents
- Track Application Status
- Verify Documents
- Approve/Reject Application
- Disburse Scholarship via DBT
- Generate Reports



Explanation of the Diagram

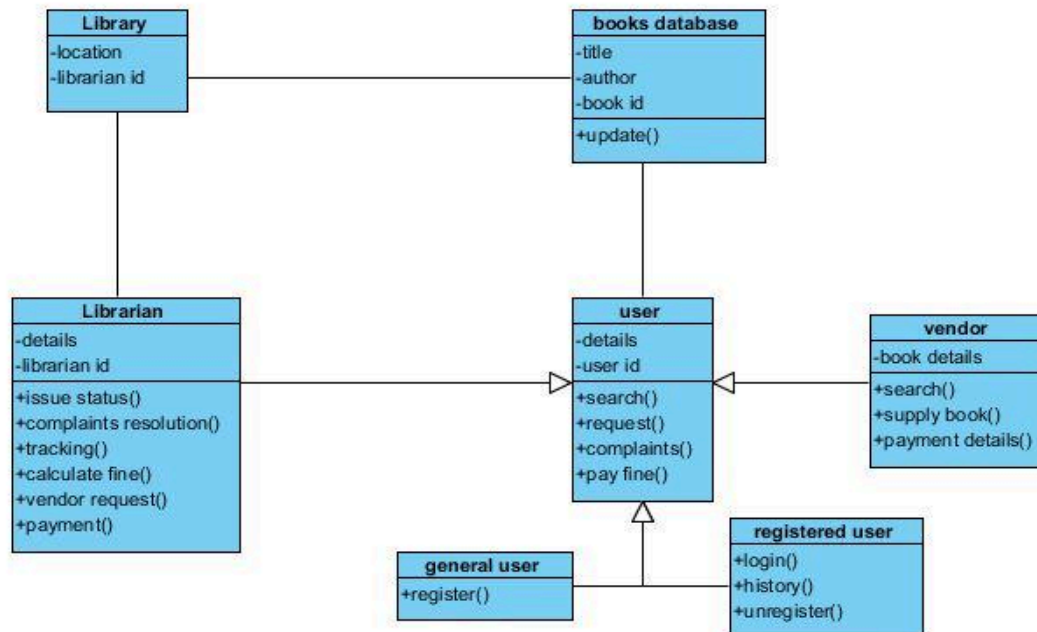
- Actors:
 - Student: Register/Login, Submit Application, Upload Documents, Track Application Status.

- Administrator: Verify Documents, Approve/Reject Application, Disburse Scholarship (DBT), Generate Reports.
- DBT System: Handles secure fund transfer.
- **Use Cases:**
 - Register/Login → includes *Receive Notifications*.
 - Submit Application → linked to Student.
 - Upload Documents → linked to Student.
 - Verify Documents → linked to Admin.
 - Approve/Reject Application → linked to Admin.
 - Disburse Scholarship (DBT) → linked to Admin and DBT System.
 - Generate Reports → linked to Admin.
 - Notifications → shared between Student and Admin.

5.2 Class Diagram

Classes:

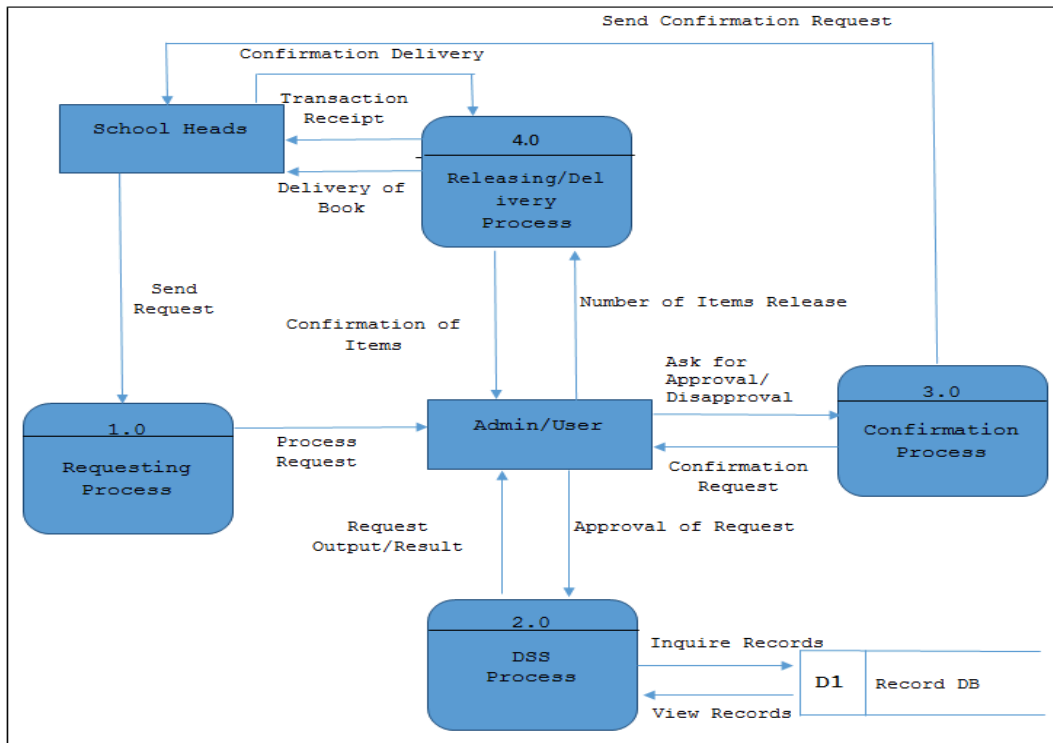
- **User** (studentId, name, email, login(), logout())
- **Application** (applicationId, status, submit(), updateStatus())
- **Document** (docId, type, upload(), verify())
- **Admin** (adminId, verifyApplication(), approveApplication(), disburseFunds())
- **DBTModule** (transactionId, transferFunds())
- **Notification** (notifyUser(), sendAlert())



5.3 Data Flow Diagram (DFD)

Level 0 (Context Diagram):

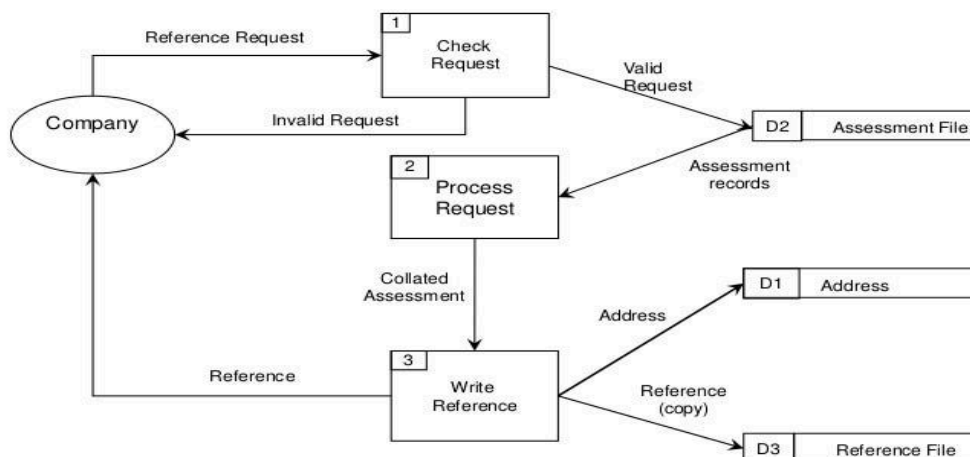
- Student → Scholarship System → Admin → DBT Gateway → Bank



Level 1 (System Decomposition):

- Registration → Application Submission → Document Upload → Verification → Approval → DBT Transfer → Notification

Level 1 DFD Example



Overview

This diagram breaks down the main system into its internal processes and data flows.

Processes:

1. **User Authentication & Registration**
 - Input: Student credentials
 - Output: Access token, user profile
2. **Application Submission**
 - Input: Scholarship form data
 - Output: Application record stored in database
3. **Document Upload & Verification**
 - Input: Uploaded documents
 - Output: Verification status
4. **Approval & DBT Processing**
 - Input: Verified application
 - Output: Fund transfer request to DBT System
5. **Notification & Reporting**
 - Input: Application status updates
 - Output: Alerts to students and reports for administrators

Data Stores:

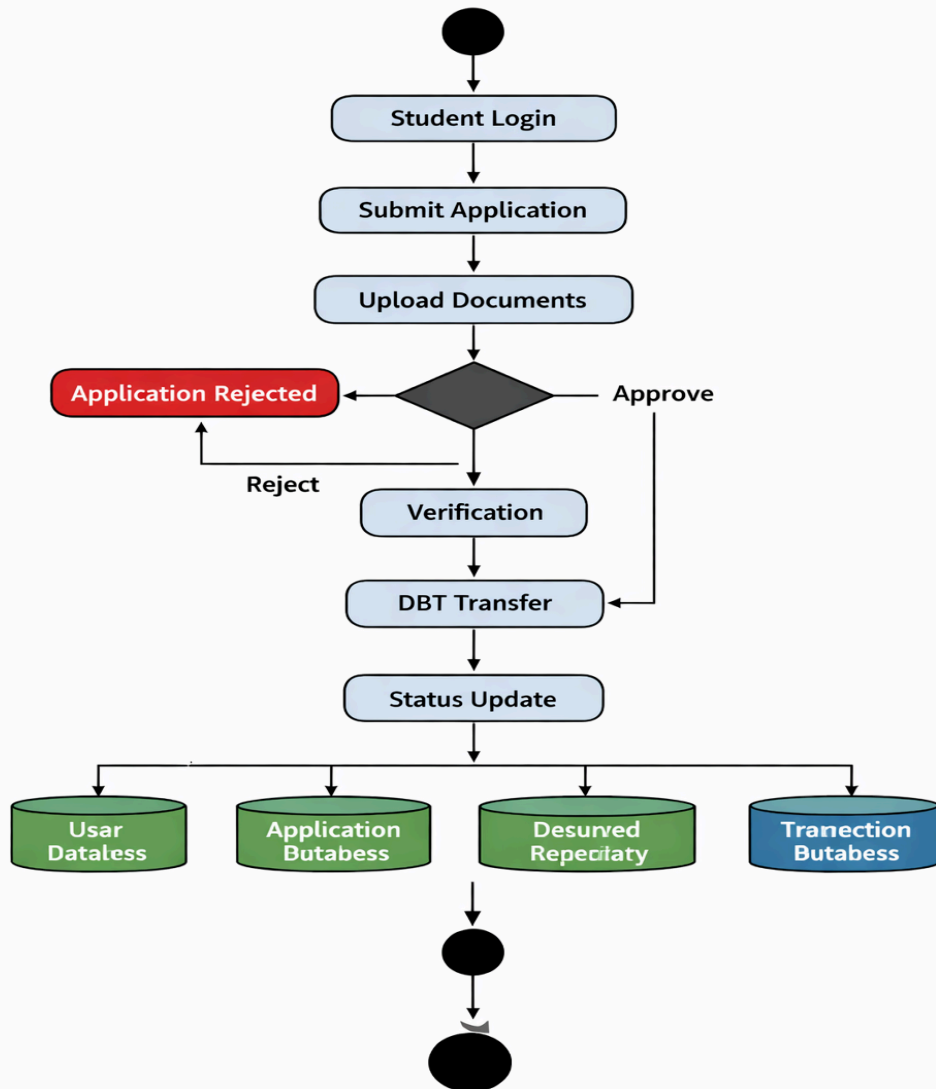
- **User Database** (student/admin info)
- **Application Database** (application details)
- **Document Repository** (uploaded files)
- **Transaction Database** (DBT records)

5.4 Sequence Diagram

Flow:

1. Student logs in → System authenticates.
2. Student submits application → System stores in DB.
3. Student uploads documents → Admin verifies.
4. Admin approves → DBT module triggers payment.
5. System sends notification → Student receives confirmation.

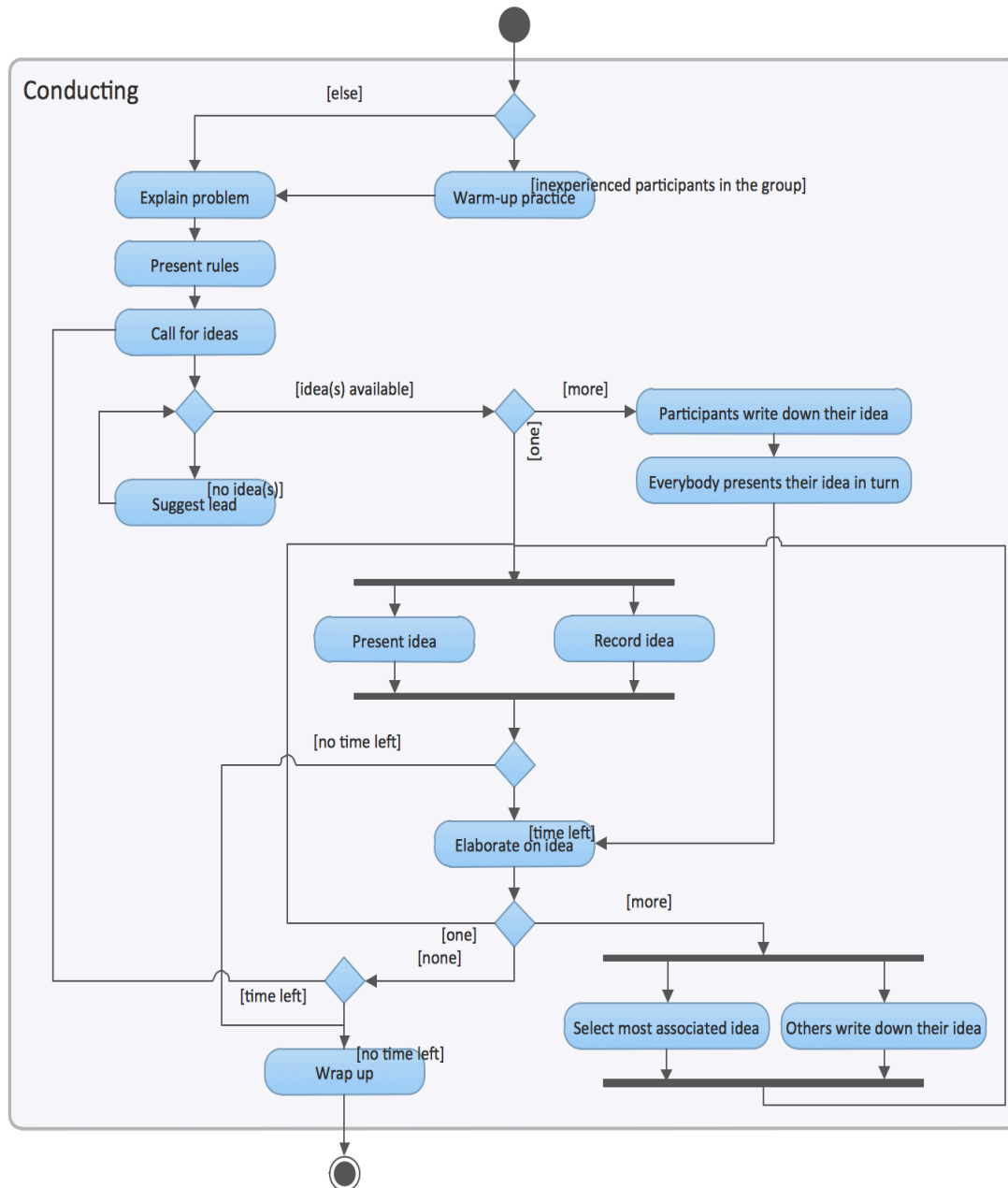
Scholarship Disbursement Process - PMSSS



5.5 Activity Diagram

Workflow:

- Student Login → Submit Application → Upload Documents → Verification → Approval → DBT Transfer → Status Update → Notification.



6. Non-Functional Requirements

- **Performance:** Handle 1000+ concurrent users.
- **Security:** JWT authentication, TLS encryption.
- **Reliability:** 99% uptime, auto-recovery for failed transactions.
- **Scalability:** Support integration with other schemes.
- **Usability:** Simple UI, responsive design.
- **Maintainability:** Modular architecture with microservices.

7. Future Requirements

- AI-based OCR for document verification
- Chatbot for student queries
- Mobile app support
- Integration with Aadhaar/Govt databases
- Advanced analytics dashboards